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RESEARCH ARTICLE

Evaluation of Minimum Reporting Limits to Determine In-Competition Use of Stimulants

Vinod S. Nair | Fatjon A. Hanelli | Chad Moore | Jenna M. Goodrum | Geoffrey D. Miller | Andre Crouch | Daniel Eichner

Sports Medicine Research and Testing Laboratory, South Jordan, Utah, USA

Correspondence: Vinod S. Nair (vnair@smrtl.org)

Received: 1 July 2024 | **Revised:** 23 August 2024 | **Accepted:** 27 August 2024

Funding: Funding for the study was provided by WADA Research Grant 23A08VN.

Keywords: in competition | minimum reporting limit | stimulants

ABSTRACT

The applicability of urinary minimum reporting limits (MRLs) to determine in-competition use of prohibited substances is an evolving topic. Most stimulants are subject to a universal MRL, despite the wide range of commercially available dosages for commonly used stimulants. Further, it is unknown whether the urinary MRL is reflective of a pharmacological dose ingested after the start of the in-competition period. To evaluate whether urinary MRLs can distinguish between in-competition and out-of-competition use, a controlled administration study was performed with three commonly used stimulants—amphetamine, methylphenidate, and modafinil at relatively low but therapeutically relevant dosages. Four to six volunteers were administered a particular drug once per day for five consecutive days. Urine, serum, dried blood spots (DBS), and oral fluid (OF) were collected during the active administration period and for 48 h after cessation of use. For all participants, urinary concentrations for all target analytes exceeded the MRL even 48 h after cessation of use. In serum and DBS, most volunteers showed detectable amounts at 48 h post use. Peak concentrations were variable between target compounds even with similar administered dosages. Further, there was a reproducible difference between serum and DBS concentrations. Interpretation of results from OF measurements was challenging due to the inability to normalize for hydration status and OF viscosity. Analyte concentrations decreased steadily over the washout period but did not correlate across matrices for all target analytes. The study reiterates the challenges associated with determining in-competition use by relying on urinary concentrations.

1 | Introduction

Stimulants have the potential for abuse as ergogenic aids during athletic competition because they temporarily improve mental and/or physical functions by increasing alertness, aggressiveness, motivation, locomotion, heart rate, and reducing fatigue [1–5]. Hence, the use of stimulants during competition is prohibited under drug class S6 of the in-competition section of the List of Prohibited Substances and Methods [6]. Even though stimulants are only prohibited in competition, they still account for ~15% of adverse findings reported worldwide [7, 8].

With a few exceptions, most stimulants are subject to a uniform minimum reporting limit (MRL) of 50 ng/mL despite the commercial availability of different dosages of prescription stimulants which are used for treatment of attention deficit disorder or narcolepsy [9]. The in-competition period generally begins at 12 a.m. on the day of competition [6]. It is possible that the use of a stimulant prior to the commencement of the in-competition period may still result in concentrations exceeding the MRL when a urine sample is collected in competition. Therefore, determination of in-competition use poses a challenge for results management. Further, urinary concentrations do not necessarily correlate with concentrations in the blood, which determine

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whether an athlete derives a performance enhancing benefit from use of the stimulant in question. While serum and dried blood spots (DBS) are now routinely used in antidoping analysis, they are not currently used to test for the presence of substances specifically prohibited in competition. Consequently, no MRL has been established in these matrices. Alternate matrices such as oral fluid (OF) that employ less invasive collection techniques have also been demonstrated to be of utility in antidoping [10–12]. OF, being a filtrate of plasma, has the potential to better reflect circulating levels than urine and may be useful in determination of in-competition use.

The aim of this study was to perform a controlled administration with commonly used prescription stimulants—amphetamine, methylphenidate, and modafinil—to determine whether urinary MRLs could differentiate between in and out-of-competition use. Further, concentrations were also measured in paired serum, DBS, and OF to determine whether one of these matrices could be better suited than urine to determine in-competition use.

2 | Material and Methods

2.1 | Collection Devices and Study Drugs

The formulations used in this study were amphetamine (Adderall XR 20 mg), modafinil (100 mg, generic), and methylphenidate (ER 18 mg, generic). Urine was collected in sterile 50-mL Falcon tubes (~10 mL per collection). OF was collected in sterile 15-mL Falcon tubes (~1 mL per collection). Capillary serum was collected from the upper arm using Tasso+ SST kits and separated by centrifugation at 1500 × *g* for 15 min. Urine, OF, and serum were stored at −20°C until analysis. DBS were collected from the upper arm using Tasso M20 kits, air dried and stored in a temperature and moisture-controlled chamber until analysis.

2.2 | Administration Study

Multidose administration occurred with the following cohorts: amphetamine—two males and two females, modafinil—four males and two females, and methylphenidate—two males and two females. Participants were between 22 and 48 years of age. Participants were instructed to take the assigned drug once per day (at ~9 a.m.) for five consecutive days. For the on-phase of administration, urine and OF were collected 6 h post administration on Days 1, 2, 3, and 4 while DBS and serum were collected 6 h post administration on Days 1 and 3. For the washout phase, all matrices were collected at 12, 24, 36, and 48 h after the last dose was administered.

For the single-dose administration, a single dose of each drug was administered to two male participants. Urine and DBS were collected at 48, 72, and 96 h post administration.

2.3 | Reagents and Chemicals

Study drugs were obtained from Jolley's Compounding Pharmacy (Sandy, UT, USA). Analytical standards were obtained from Sigma Aldrich (modafinil, methylphenidate, and ritalinic acid), Toronto Research Chemicals (modafinil acid), and Cerilliant (amphetamine). Internal standards (ISs) were obtained from Cerilliant (amphetamine-d5, ritalinic acid-d4, modafinil-d10, and methylphenidate-d9). Methanol and acetonitrile were purchased from Honeywell. Formic acid, hydrochloric acid, and ammonium formate were purchased from Sigma Aldrich.

2.4 | Sample Preparation

The IS mix for urine was comprised of amphetamine-d5, ritalinic acid-d4, modafinil-d10, and methylphenidate-d9 at 0.5 ng/μL. For the other matrices, the IS mix was comprised of the same compounds but at a 10-fold lower concentration. For urine, a 50-μL aliquot was fortified with 20 μL of IS solution. Five hundred microliter of 0.1% formic acid in water was added, and the plate was transferred to the autosampler. For serum, a 50-μL aliquot was fortified with 20-μL IS. Six hundred microliter of cold acetonitrile was added, and samples were shaken at 600 rpm for 10 min. The samples were then centrifuged at 4500 × *g* for 5 min, and 500 μL of the supernatant was transferred into a new 96-well plate and acidified with 50 μL of 0.1-N hydrochloric acid in methanol and dried under a gentle stream of nitrogen. The samples were then reconstituted in 150 μL of 0.1% formic acid in water and transferred to the autosampler. For DBS, a single spot was analyzed for all time points except one, where all four spots were extracted to assess reproducibility. Spots were punched out into a 96-well plate, and 20 μL of IS was added, followed by 600 μL of methanol. The plate was then sonicated for 15 min and shaken at 1000 rpm for another 20 min. The plate was then centrifuged at 4500 × *g* for 5 min, 500 μL of the supernatant was transferred to a new plate and acidified, dried, and reconstituted as described above. For OF, a 50-μL aliquot was fortified with 20-μL IS, and 600 μL of methanol was added. The plate was then shaken at 600 rpm for 10 min, followed by centrifugation at 4500 × *g* for 5 min. Three hundred microliter of the supernatant was acidified, dried, and reconstituted as described above.

2.5 | Instrumentation

Analysis was performed on a Waters Xevo T-XS mass spectrometer coupled with an Acquity I Class UPLC and autosampler system (Waters Corporation, Milford, MA, USA). Chromatographic separation was achieved with a Kinetex C18 column (2.1 × 50 mm, 2.6 μm; Phenomenex Corporation, Torrance, CA, USA). The mobile phases were 0.1% formic acid in water containing 10-mM ammonium formate (A) and methanol (B). The gradient was held at 1% B for 1 min, increased to 40% B over the next 6 min and held for 1 min, then increased to 95% B over the next 2 min and held there for 1 min. The system was then returned to starting conditions and equilibrated for 2 min. The capillary voltage was set at 3 kV. Desolvation was achieved at 250°C with a gas flow rate of 600 L/h. The MRMs were acquired in positive mode.

2.6 | Validation

The primary target compounds for the analysis were amphetamine, ritalinic acid, and modafinil acid. In addition, the parent compounds (methylphenidate and modafinil) were also

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TABLE 1 | Summary of validation.

Matrix	Compound	Linear range (ng/mL)	LoQ	Reproducibility (interday)		Robustness (interday)		Recovery
				Accuracy (%)	Imprecision (%RSD)	Accuracy (%)	Imprecision (%RSD)	
Urine	Modafinil	25–2000	25	109	4.3	99	9.2	N/A
	Modafinil acid	25–2000	50	111	4.6	101	15.9	
	Methylphenidate	2.5–200	2.5	111	2.9	105	6.5	
	Ritalinic acid	25–2000	25	105	2.8	99	8	
	Amphetamine	25–2000	25	107	3.1	103	7	
Serum	Modafinil	10–1000	10	103	1.7	108	14.8	93%
	Modafinil acid	10–1000	10	114	7.1	110	13	85%
	Methylphenidate	0.1–10	0.1	107	6.8	97	16.2	94%
	Ritalinic acid	1–100	1	105	7.9	100	16.8	80%
	Amphetamine	1–100	1	98	4.2	98	12.9	93%
DBS	Modafinil	10–1000	1	100	11.1	106	11.4	93%
	Modafinil acid	10–1000	20	100	11.6	107	13.6	66%
	Methylphenidate	0.1–10	0.1	99	6.2	115	12	80%
	Ritalinic acid	1–100	1	100	5.8	108	14.2	71%
	Amphetamine	1–100	1	100	15.4	107	12.3	76%
OF	Modafinil	10–1000	1	100	2.14	87	14.4	ND
	Modafinil acid	10–1000	20	100	3.22	86	17.2	
	Methylphenidate	0.1–10	0.1	100.1	3.22	82	12.5	
	Ritalinic acid	1–100	1	100	2.62	81	10.7	
	Amphetamine	1–100	1	100	2.25	102	18.2	

Note: Between subject variability was very large for OF; hence, a recovery estimate was not attempted. As such, while concentrations have been measured for OF, normalization of concentrations was not possible; hence, there are large differences between subjects. Interpretation of measured values is challenging and utility for the purposes of this study appears to be limited.

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evaluated. Concentrations were estimated using a calibration curve that was extracted and analyzed with each analytical run. Intraday and interday reproducibility were assessed by analyzing a single matrix ($n = 15$) at the lower limit of quantitation (LLoQ) over 3 days. Robustness was assessed by analyzing 10 different matrices at the LLoQ over 2 days. Recovery was established by comparing responses of pre and post extraction spikes. Carryover was evaluated by analyzing blank samples after the highest point of the calibration curve. The validation results are summarized in Table 1.

3 | Results and Discussion

3.1 | Urine

Urine results for the multidose administration are illustrated in Figure 1. Measured concentrations (adjusted for specific gravity) were plotted on a log scale to permit visualization of the MRL in the same plot. For the multidose cohort, at 48 h after cessation of use, concentrations for all participants were above the MRL of 50 ng/mL when considering target analytes (amphetamine: 291–414 ng/mL, modafinil acid: 513–5589 ng/mL, and ritalinic acid: 348–641 ng/mL). In addition, the parent compound modafinil was also above 50 ng/mL (126–328 ng/mL) at 48 h post use. The parent compound methylphenidate was the exception and fell below 50 ng/mL approximately 24 h after the last dose. There was no noticeable accumulation over the dosing period. Results from the single-dose administration, where samples were collected 96 h post administration, are presented in Table 2. The concentration of modafinil parent dropped below 50 ng/mL by the last collection, but modafinil acid was still above 50 ng/mL, reflecting the much larger dosage strength of modafinil compared to the other two compounds administered. Urinary amphetamine concentrations declined below 50 ng/mL at the 72-h collection. Metabolites of amphetamine were not considered in this analysis because the parent compound itself is the target in antidoping analysis.

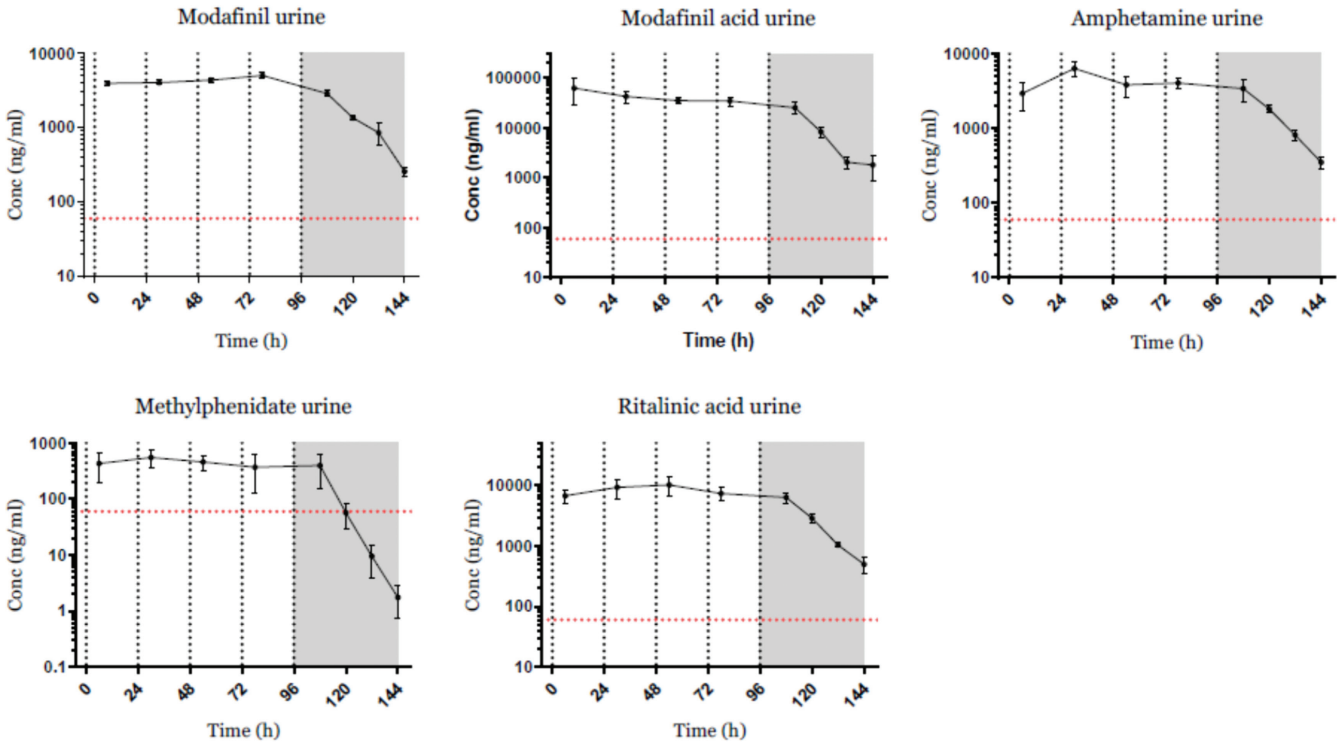

FIGURE 1 | Ranges of urinary concentrations (adjusted for specific gravity) for administered stimulants and target metabolites (where applicable). The dotted lines indicate times of administration. Shaded area indicates the washout period.

TABLE 2 | Urinary concentrations (adjusted for SG) and DBS concentrations from the single-dose administration at 72 and 96 h post use.

	Urine concentrations (ng/mL)		DBS concentrations (ng/mL)	
Compound	**72 h**	**96 h**	**72 h**	**96 h**
Modafinil	43–85	2–16	20–46	7–11
Modafinil acid	232–460	98–277	—	—
Methylphenidate	ND	ND	ND	ND
Ritalinic acid	67–354	20–106	7.5–50	2.6–21
Amphetamine	6–7.5	ND	ND	ND

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Ritalinic acid was below the MRL for one subject and above for the other subject at 96 h.

3.2 | Serum

Amphetamine, ritalinic acid, and the parent compound modafinil were all detectable above the method LoQ at 48 h post use. Average concentrations at the 48-h collection were 2.9 (amphetamine), 19.5 (ritalinic acid), and 87 ng/mL (modafinil). Results are shown in Figure 2. However, in contrast with urine, modafinil acid concentrations declined rapidly in serum and dropped below the method LoQ at 24 h post use. As observed in urine, methylphenidate parent concentrations declined rapidly and fell below the method LoQ at about 24 h post use. Though methylphenidate and amphetamine were dosed in similar amounts, the serum concentrations for methylphenidate parent were approximately 10-fold lower than amphetamine; however, ritalinic acid concentrations were approximately 10-fold higher than amphetamine. These results are in line with the manufacturer provided pharmacokinetics [13, 14]. Amphetamine is known to produce other metabolites (e.g., *p*-hydroxyamphetamine) that were not monitored in this study, which likely explains the discrepancy in observed serum concentrations above [15].

3.3 | DBS

Results for DBS testing are also illustrated in Figure 2 along with serum results. For the multiday cohort, DBS concentrations showed the same trend as serum concentrations; however, the observed concentrations in DBS were noticeably and reproducibly higher than in serum, except for modafinil acid. Average concentrations at the 48-h collections were 3.8 (amphetamine), 35 (ritalinic acid), and 110 ng/mL (modafinil). To evaluate the retention of stimulants in the red blood cell fraction, a single dose of amphetamine was administered to one subject, and whole blood was collected at 6 h post administration. The plasma and red blood cells were separated by centrifugation, and both fractions were evaluated for presence of amphetamine. Amphetamine was detected in the extracts from both fractions, which may explain the higher concentrations detected in DBS as compared to serum. Considering that the other study compounds show similar differences between the serum and DBS concentrations, it is likely that a similar phenomenon occurs with those compounds also. It should be noted that this evaluation was only performed on a qualitative basis to examine whether some fraction of the drug was bound to red blood cells. However, there is no evidence that any of these compounds accumulate in the red blood cells because they do not exhibit the prolonged detection windows seen with compounds like dorzolamide or meldonium that have indeed been demonstrated to be sequestered in the red blood cells [16].

Results from the single-dose administration with collections up to 96 h are summarized in Table 2. Amphetamine was no longer detectable at 72 h, but modafinil parent and ritalinic acid were still detectable up to 96 h post administration.

The variability in measured concentration from spot to spot was also evaluated. A time point with concentrations in the middle of the curve was selected for each stimulant (modafinil—24 h post use, ritalinic acid—24 h post use, and amphetamine—6 h post use) and all four spots from the Tasso M20 were extracted and analyzed. Most subjects showed low variability in

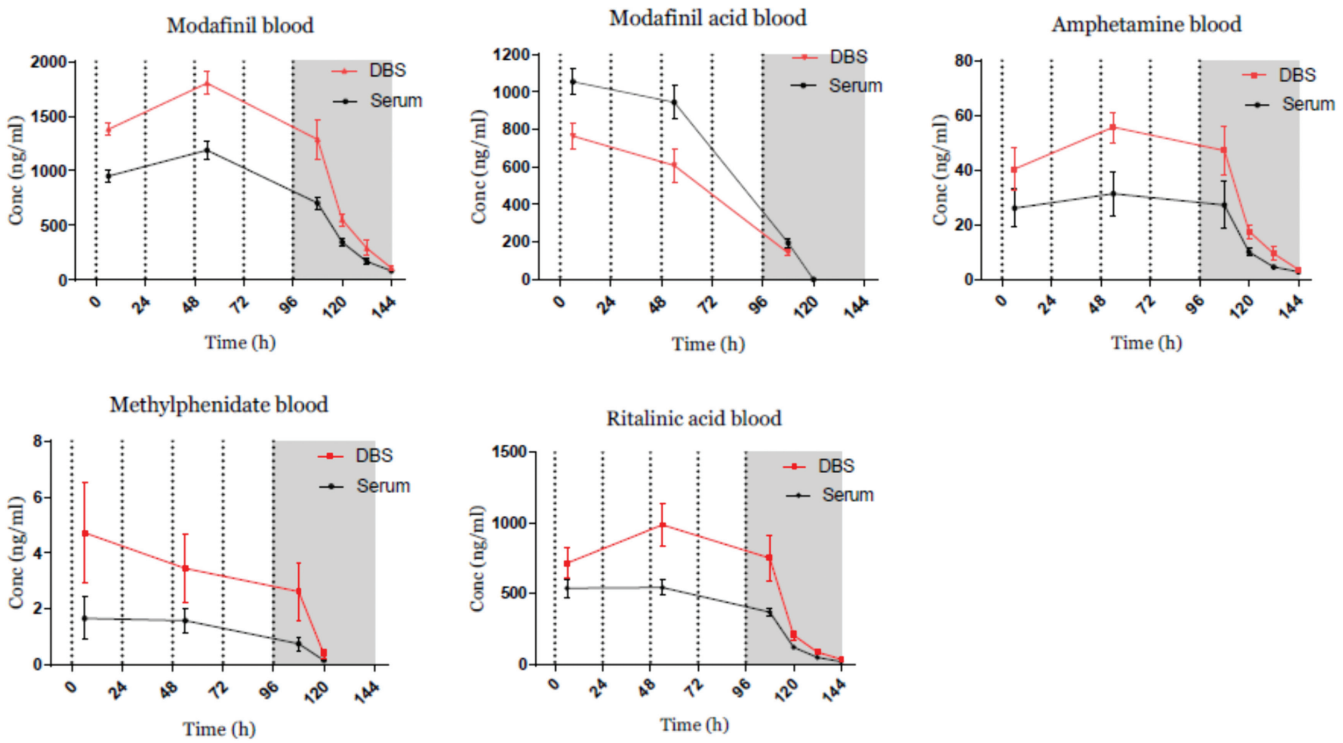

FIGURE 2 | Ranges of serum and DBS concentrations for administered stimulants and target metabolites (where applicable). The dotted lines indicate times of administration. Shaded area indicates the washout period.

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concentrations between the four spots. Results are presented in Table 3. Two subjects (S9 for modafinil and S16 for amphetamine) showed particularly high variability, likely related to overfilling of the cartridge. In these two collections, the cartridges had been grossly overfilled such that a layer of blood had caked over the bottom two spots, leading to high spot to spot variability. It is therefore advisable when using the Tasso M20 devices to disengage the device from the arm once all four spots have been saturated and blood is visible at the base of the cartridge.

TABLE 3 | Spot to spot variability from a single DBS collection in the Tasso M20 device.

	Subject	Average (ng/mL)	% CV
Modafinil	S1	567	7
	S2	554	7.1
	S7	528	4
	S9	707	27.1*
	S10	363	8.4
	S18	386	3.8
Amphetamine	S6	36	8.7
	S11	60	9.3
	S16	87	35.2*
	S19	54	2.8
Ritalinic acid	S8	188	2.5
	S14	238	14.1
	S20	145	8.2

Note: The high variability in the two instances indicated with an asterisk is likely due to overfilling of the device.

3.4 | OF

Results for OF were generally similar to serum, although with much larger intersubject variability. Modafinil acid could not be detected in any of the subjects, but as seen in serum and DBS, modafinil parent, amphetamine, and ritalinic acid could be detected in all subjects at 48 h post use. Results are shown in Figure 3. Methylphenidate parent concentrations were low throughout the administration and dropped below the method LoQ at 24 h post use, mirroring the trends seen in serum and DBS. Intersubject variability was higher than in the other matrices, possibly due to pH effects affecting partitioning into OF and also the inability to normalize for viscosity and specific gravity.

Based on the above results, it does not appear that a urinary MRL of 50 ng/mL is diagnostic enough to distinguish between in and out-of-competition use. For the prescription stimulants evaluated in this study, an in-competition test could produce an adverse finding even when the stimulant in question were used during the out-of-competition period (within 48–96 h of the test). It should also be noted that this study used relatively

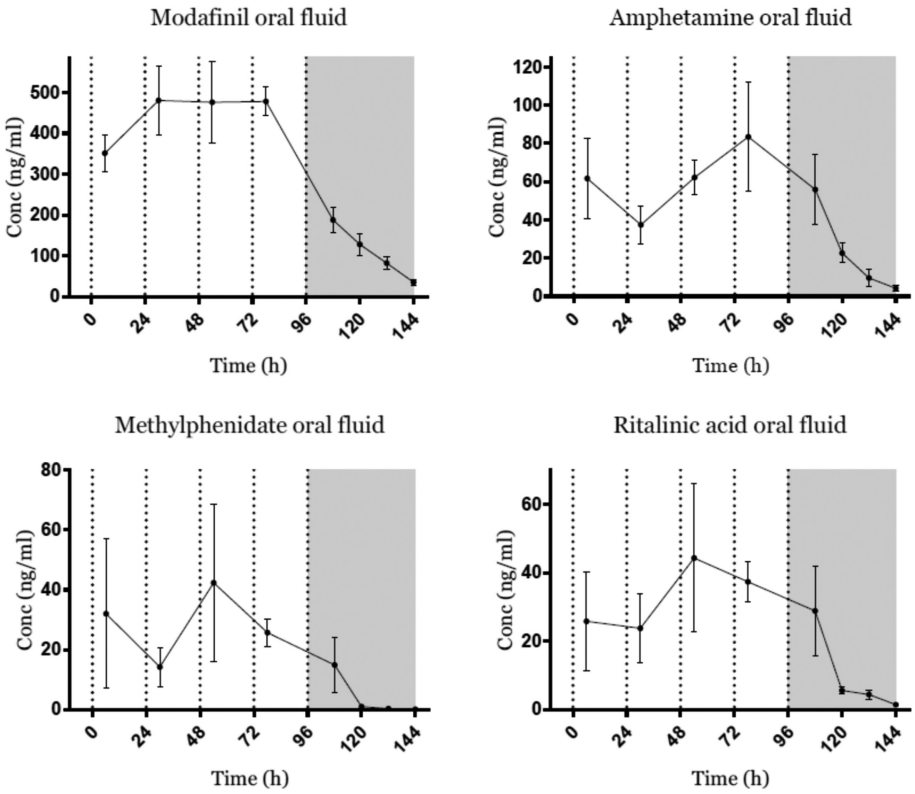

FIGURE 3 | Ranges of OF concentrations for administered stimulants and target metabolites (where applicable). The dotted lines indicate times of administration. Shaded area indicates the washout period.

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low strengths of available prescription doses. Use of higher strengths could further increase the detection window for concentrations above the MRL. Alternate matrices were also not useful in determining time of consumption as all substances evaluated were detectable at the end of the sampling period. This is consistent with published reports where the windows of detection in serum after self-reported stimulant usage were as high as 77.5 h [15]. While serum, DBS, or OF provide a more realistic view of systemic presence, it is difficult to propose a cutoff concentration that can conclusively distinguish between use during the in-competition period or use prior to it. However, if detection of systemic presence above a certain concentration can be correlated with a performance enhancing effect, then it may be possible to determine a DBS or serum MRL. Detection above this level may indicate that the athlete was deriving performance enhancement from the stimulant in question.

The fact that stimulants comprise ~15% of adverse findings reported in ADAMS indicates that use of stimulants in the athlete population is not negligible. On the other hand, a 2-year study involving 11 WADA-accredited laboratories indicated that out-of-competition use of stimulants was not particularly prevalent [17]. However, the above study was published in 2011, and the landscape of stimulant prescriptions appears to have changed since then. For instance, evaluation of the Massachusetts Prescription Drug Monitoring Program indicated a 70% increase in stimulant prescriptions from 2011 to 2021 [18]. It is conceivable that athletes may be medically prescribed a stimulant but fail to complete the necessary Therapeutic Use Exemption (TUE) requirements and therefore produce an adverse finding in competition. The greater availability of prescription stimulants may also contribute to more accessible recreational use out of competition without the intent to derive a performance enhancing effect. The relatively long detection window established in this study might then lead to an inadvertent adverse finding in competition. Athletes subject to antidoping testing should be cautioned about the use of stimulants around competition events without a valid TUE. To enable better results management, it may be worth considering whether commonly prescribed stimulants should be prohibited at all times and the TUE process should be invoked if medical use is necessary. Alternatively, compound specific MRLs could be introduced for these substances as enforced for glucocorticoids [9].

4 | Conclusions

Relatively low doses of prescription stimulants can produce urinary concentrations in excess of the MRL if used within 48–96 h of sample collection. The use of alternative matrices such as serum or DBS can provide a more realistic insight into systemic concentrations but still does not allow distinction between in-competition and out-of-competition use. Interpretation of OF concentrations is challenging due to the inability to normalize measured values. Despite ease of collection, OF is therefore not well suited as a matrix for determination of in-competition use. Finally, a distinct set of MRLs or a change in prohibited status may be warranted for commonly prescribed stimulants to facilitate reasonable results management.

Acknowledgements

The authors would like to thank Xander LaFleur for specimen organization and Dixie Peeler for clinical support. Funding for the study was provided by WADA Research Grant 23A08VN.

Conflicts of Interest

The authors declare no conflicts of interest.

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